

Ex. 26. Pursuit on a sphere

1) In a local coordinate system, the state equations of the robot are:

$$\begin{cases} \dot{x} = \cos \psi \\ \dot{y} = \sin \psi \\ \dot{\psi} = u \end{cases}$$

But we also know that a first-order approximation is also:

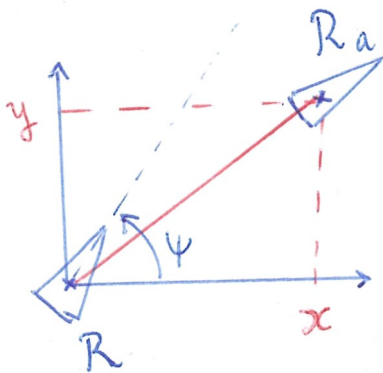
$$\begin{cases} x = \rho \cdot \cos(l_y) \cdot l_x \\ y = \rho \cdot l_y \\ z = \rho \end{cases} \quad \text{so} \quad \begin{cases} \dot{x}_{ldx} = \rho \cdot \cos(l_y) \cdot \dot{l}_x \\ \dot{y}_{ldy} = \rho \cdot \dot{l}_y \\ \dot{z}_{ld\rho} = \dot{\rho} \end{cases}$$

Then we deduce:

$$\begin{cases} \cos \psi = \rho \cdot \cos(l_y) \cdot \dot{l}_x \\ \sin \psi = \rho \cdot \dot{l}_y \\ \dot{\psi} = u \end{cases} \quad \Leftrightarrow$$

$$\begin{cases} \dot{l}_x = \frac{\cos \psi}{\rho \cdot \cos l_y} \\ \dot{l}_y = \frac{\sin \psi}{\rho} \\ \dot{\psi} = u \end{cases}$$

3)



If m is the point where the robot R is, and a the one where R_a is, we consider: $\vec{ma} = \begin{pmatrix} x \\ y \end{pmatrix}$

The orientation of the robot R is given by:

$$\vec{m} = \begin{pmatrix} \cos \psi \\ \sin \psi \end{pmatrix}$$

R follows R_a if the previous vectors are collinear.

So a simple control law could be:

$$u = k \cdot \det(\vec{m}, \vec{ma})$$

with k a constant.

Remarks

In simulation, the control law seems to work correctly, but some irregularities appear on the top position of the earth. It is when $l_y = \frac{\pi}{2} \Rightarrow \cos l_y = 0 \Rightarrow \dot{l}_x \rightarrow \infty$

Ex. 27. Artificial potentials

$$1) V(p) = -\hat{v}^T \cdot p + \|p - \hat{p}\|^2 + \frac{1}{\|p - \hat{q}\|}$$

$$\begin{aligned} w(p, t) &= -\text{grad } V(p) \\ &= -\text{grad}(-\hat{v}^T \cdot p) - \text{grad}(\|p - \hat{p}\|^2) - \text{grad}\left(\frac{1}{\|p - \hat{q}\|}\right) \end{aligned}$$

$$w(p, t) = \hat{v} - 2(p - \hat{p}) + \frac{(p - \hat{q})}{\|p - \hat{q}\|^3}$$

2) The first block tells us $\begin{cases} \dot{v} = u_1 \\ \dot{\theta} = u_2 \end{cases}$ where u_1, u_2 are the control variables.

We need v and θ to follow $\bar{v}, \bar{\theta}$ with a proportional control.

Then, we could use:

$$\begin{cases} u_1 = k_1 (\bar{v} - v) \\ u_2 = k_2 \text{sawtooth}(\bar{\theta} - \theta) \end{cases} \text{ with } k_1, k_2 \in \mathbb{R}^+$$

$\bar{v}$ and $\bar{\theta}$ are given by the vector field setpoint $w(p, t)$:

$$\bar{v} = \|w(p, t)\|$$

$$\bar{\theta} = \text{angle}(w(p, t)) = \arctan\left(\frac{w_y(p, t)}{w_x(p, t)}\right) \text{ where } w(p, t) = \begin{pmatrix} w_x(p, t) \\ w_y(p, t) \end{pmatrix}$$

Ex. 28. Flocking

The three rules of Reynolds: alignment, cohesion and separation can be made with the same kind of potential method as ex. 27.

For each robot i , we have

$$w_i = k_1 \sum_{j \neq i} v_j^0 - 2 \cdot k_2 \sum_{j \neq i} (p_i - p_j) + k_3 \sum_{j \neq i} \frac{p_i - p_j}{\|p_i - p_j\|^3}$$

where k_1, k_2, k_3 are some positive constant chosen to modulate the proportion of each part of Reynolds rules.

Then can either modulate each part with k_1, k_2, k_3 , or normalize each part by dividing by its norm.

Then, we have $u_2 = \text{sawtooth}(\bar{\theta}_i - \theta_i)$ with $\bar{\theta}_i = \arctan\left(\frac{w_{y_i}}{w_{x_i}}\right)$